How-To Library

Informatica

Configuring Optimized Spark Mode for PowerExchange for Google BigQuery Data Objects

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Abstract

This article describes how to configure the Optimized Spark mode to increase the performance of PowerExchange for Google BigQuery to read data from or write data to Google BigQuery from Informatica Developer. Learn how to configure the Google BigQuery Data Object Read and Write operation properties when you use Optimized Spark mode.

Supported Versions

- PowerExchange for Google BigQuery 10.4.0

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Overview

You can configure PowerExchange for Google BigQuery to use Optimized Spark mode and read data from or write data to Google BigQuery when you use the Developer tool and run a mapping on the Spark engine.

To configure Optimized Spark mode, perform the following tasks:

1. Complete the following prerequisite tasks:
   a. Install the Informatica services 10.4.0.
   b. Install the Developer tool.
2. Create a Google BigQuery connection and select **Simple** as the connection mode.
3. Create a Google BigQuery data object and the data object read and write operations.
4. Create a mapping and run the mapping on the Spark engine.

Configure the Google BigQuery Data Object Read Operation Properties

When you select **Optimized** as the Spark mode in a Google BigQuery data object read operation, you can configure the following properties:

Property	Description
Source Dataset ID	Optional. Overrides the Google BigQuery dataset name specified in the Google BigQuery data object read operation.
Source Table Name	Overrides the Google BigQuery table name specified in the Google BigQuery data object read operation.
Number of Rows to Read	Specifies the number of rows to read from the Google BigQuery source. Enter a number greater than 0.

Property	Description
Spark Mode	Applies when you run a mapping on the Spark engine. Specifies the mode of the Spark engine to read data from the Google BigQuery source. Select **Optimized**.
Read Mode	Specifies the read mode to read data from the Google BigQuery source. Select Direct. In direct mode, PowerExchange for Google BigQuery reads data directly from the Google BigQuery source table. **Note:** Staging mode is not applicable for Optimized Spark mode.
Partitions Per Core	Specifies the number of partitions to split the data. Default is 1.
Allow Large Results	Not applicable for Optimized Spark mode.
Query Results Table Name	Not applicable for Optimized Spark mode.
Job Poll Interval in Seconds	Not applicable for Optimized Spark mode.
Number of Threads for Downloading Staging Files	Not applicable for Optimized Spark mode.
Data Format of the staging file	Not applicable for Optimized Spark mode.
Local Stage File Directory	Not applicable for Optimized Spark mode.
Staging File Name	Not applicable for Optimized Spark mode.
Enable Staging File Compression	Not applicable for Optimized Spark mode.
Persist Destination Table	Not applicable for Optimized Spark mode.
pre SQL	Not applicable for Optimized Spark mode.
pre SQL Configuration	Not applicable for Optimized Spark mode.
post SQL	Not applicable for Optimized Spark mode.
post SQL Configuration	Not applicable for Optimized Spark mode.
SQL Override Query	Not applicable for Optimized Spark mode.
Use Legacy SQL for SQL Override	Not applicable for Optimized Spark mode.

Configure the Google BigQuery Data Object Write Operation Properties

When you select Optimized as the Spark mode in a Google BigQuery data object write operation, you can configure the following properties:

Property	Description
UpdateMode	Not applicable for Optimized Spark mode or Generic Spark mode.
Target Dataset ID	Optional. Overrides the Google BigQuery dataset name specified in the Google BigQuery data object write operation.
Target Table Name	Optional. Overrides the Google BigQuery target table name specified in the Google BigQuery data object write operation.
Spark Mode	Applies when you run a mapping on the Spark engine. Specifies the mode of the Spark engine to write data into the Google bigQuery target. Default is Generic mode. Select **Optimized**.
Write Mode	Specifies the mode to write data to the Google BigQuery target. Select **Bulk**. In bulk mode, PowerExchange for Google BigQuery first writes the data to a staging file in Google Cloud Storage. **Note:** Streaming mode is not applicable for Optimized Spark mode.
Create Disposition	Specifies whether PowerExchange for Google BigQuery must create the target table if it does not exist. Select Create never. If the table does not exist, PowerExchange for Google BigQuery does not create the table. **Note:** Create if needed is not applicable for Optimized Spark mode.
Write Disposition	Specifies how PowerExchange for Google BigQuery must write data in bulk mode if the target table already exists. You can select one of the following values: - Write append. If the target table exists, PowerExchange for Google BigQuery appends the data to the existing data in the table. - Write truncate. If the target table exists, PowerExchange for Google BigQuery overwrites the existing data in the table. - Write empty. If the target table exists and contains data, PowerExchange for Google BigQuery displays an error and does not write the data to the target. PowerExchange for Google BigQuery writes the data to the target only if the target table does not contain any data. **Note:** Write disposition is applicable for bulk mode. Write disposition is applicable only when you perform an insert operation on a Google BigQuery target.
Data Format of the staging file	Specifies the data format of the staging file. Select JSON (Newline Delimited). **Note:** CSV format is not applicable for Optimized Spark mode.
Persist Staging File After Loading	Indicates whether PowerExchange for Google BigQuery must persist the staging file in the Google Cloud Storage after it writes the data to the Google BigQuery target. You can persist the staging file if you want to archive the data for future reference. By default, PowerExchange for Google BigQuery deletes the staging file in Google Cloud Storage. This property applies to bulk mode.

Property	Description
Staging File Name	Not applicable for Optimized Spark mode.
Enable Staging File Compression	Not applicable for Optimized Spark mode.
Job Poll Interval in Seconds	Not applicable for Optimized Spark mode.
Number of Threads for Uploading Staging file	Not applicable for Optimized Spark mode.
Local Stage File Directory	Not applicable for Optimized Spark mode.
Allow Quoted Newlines	Not applicable for Optimized Spark mode.
Field Delimiter	Not applicable for Optimized Spark mode.
Allow Jagged Rows	Not applicable for Optimized Spark mode.
Pre SQL	Not applicable for Optimized Spark mode.
Pre SQL Configuration	Not applicable for Optimized Spark mode.
Post SQL	Not applicable for Optimized Spark mode.
Post SQL Configuration	Not applicable for Optimized Spark mode.

General Guidelines to Configure Optimized Spark Mode

Consider the following best practices when you configure Optimized Spark mode to read data from or write data to Google BigQuery:

- You must not configure a Google BigQuery connection in Hybrid or Complex connection mode.
- You must not configure a Native expression to query the Google BigQuery tables.
- In the data object write operation advanced properties, you must not configure the **Create Disposition** property to **Create if needed**.
- In the data object write operation advanced properties if you select **CSV** as the **Data Format of the staging file**, you must not select the **Persist Staging File After Loading** property.
- In the data object write operation advanced properties, you must not
- When you read data from Google BigQuery, you must ensure that the Google BigQuery data object must not contain columns of the following data types:
  - Date
  - Time
  - Timestamp
- When you read data from Google BigQuery, you must ensure that the Google BigQuery data object must not contain repeated columns.
- When you write data to Google BigQuery, you must ensure that the Google BigQuery data object must not contain columns of the Record data type and repeated columns.

- When you specify a Google Cloud Storage path in the Google BigQuery connection, PowerExchange for Google BigQuery creates a folder in the specified path with the following format:
  `<project_id>:<dataset_id>.<table_name>_<timestamp>_<unique identifier>`

  When you write data to a Google BigQuery table in bulk mode, PowerExchange for Google BigQuery first writes the data to the staging files available in the same folder.

Author

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